Data Science

Lecture 01: Introduction

Prof. Dr. Michael Granitzer

Module 1

The Data Deluge

> ***Guiding Question:*** *We generate more data than we can store. Does having more of it make us know more?*

How Much Data Exists?

Year	Global datasphere	Per person on earth
2002	5 exabytes of new stored information (Lyman & Varian 2003)	≈ 0.8 GB
2010	≈ 1.2 zettabytes (Gantz & Reinsel 2010)	≈ 170 GB
2020	≈ 64 ZB; IDC's 2012 *forecast* had said 40 ZB (Gantz & Reinsel 2012)	≈ 8,000 GB
2025	≈ 175 ZB — a 2018 *forecast* (Reinsel et al. 2018)	≈ 21,000 GB

In 2007 the amount of information produced first **surpassed available storage capacity**. Since then, most data that is generated is never stored.

Growth Is Not the Interesting Part

Volume

More bytes than anyone can read, index or curate.

Heterogeneity

Text, sensors, images, logs, transactions — rarely in a common schema.

Velocity

Streams that must be acted on before they can be stored.

> The scarce resource is not data. It is **the ability to turn data into justified conclusions** — which is what this course is about.

Science at Scale

Large Hadron Collider

1 MB per event, 600 million events per second — roughly **18 zettabytes per year** if everything were kept.

A single tier-1 centre stores ~2.2 petabytes per year.

The filter is the experiment

Almost all events are **discarded in hardware**, in real time, by trigger rules written in advance.

What is kept is decided by a theory of what matters.

Data at Web Scale

Search

Web-scale indexing processes petabytes per day and answers in milliseconds.

Platforms

Interaction logs — clicks, dwell time, scrolls — dwarf the content itself.

Models

Training corpora for language models are measured in trillions of tokens.

In every case the value comes from **aggregation across many weak signals**, not from any single record.

Data is never raw: at the LHC a physics theory decides what is recorded, and on the web value comes from aggregating signals that are individually worthless. **Someone always chose what the dataset contains.**

Module 2

Does Data Speak for Itself?

> ***Guiding Question:*** *Twenty-nine teams analysed the same dataset and the same question. Why did they not get the same answer?*

"The End of Theory"

> *With enough data, the numbers speak for themselves. […] Correlation supersedes causation, and science can advance even without coherent models, unified theories, or really any mechanistic explanation at all.*

Chris Anderson, *Wired*, 2008 (Anderson 2008). The argument has returned, almost verbatim, in claims that sufficiently large models make domain theory unnecessary.

Is it right?

One Dataset, Many Answers

Twenty-nine research teams received **the same dataset** and the same question:

> *Are football referees more likely to give red cards to dark-skinned players than to light-skinned players?*

The result

Odds ratios ranged from **0.89 to 2.93**. Twenty teams found a significant effect; nine did not.

The reason

Not mistakes: the spread persisted among analyses peer-rated as sound. Teams made **different defensible choices** — covariates, exclusions, model.

The data did not determine the answer. The **analysis** did. (Silberzahn et al. 2018)

Correlation Is Not Causation

Spurious correlations are easy to find and get **easier as datasets grow wider**: with enough variables, some pair will track another closely by chance alone.

More columns: chance

Testing many pairs makes an extreme correlation likely *somewhere*, even when every pair is independent.

More rows: no cure for confounding

A confounded correlation does not go away with more rows. The interval shrinks around a real but **non-causal** number: more precise, not more valid.

Summary Statistics Hide the Data

Datasets whose **summary numbers agree** to two decimal places — same averages, same spread, same strength of association — can look completely different when plotted: a dinosaur, a star, a set of lines (Matejka & Fitzmaurice 2017).

We will make "average", "spread" and "association" precise in Lecture 02, and meet this phenomenon again there as Anscombe's quartet.

> Plot the data. Then plot it again, differently.

Where This Leaves Us

Data does not speak for itself

Every dataset was selected, filtered and encoded by someone.

Data cannot be read intuitively

Human intuition fails at scale, in high dimensions, and with base rates.

So theory is needed *more*

Models make the assumptions explicit enough to be criticised.

Big data does not end theory. It **raises the cost of implicit theory**, because assumptions that stay unstated now get automated and scaled.

Module 3

What Is Data Science?

> ***Guiding Question:*** *If data science is not statistics, not machine learning and not engineering, what is left that is its own?*

A Definition

> *Data science is the study of the generalizable extraction of knowledge from data. — Vasant Dhar (Dhar 2013)*

> **Course definition, after Dhar.** Data science researches and applies technologies, systems and algorithms to gain knowledge from potentially large, heterogeneous and dynamically changing data sets — for a particular domain.

The phrase doing the work is **"for a particular domain"**: without it, there is no criterion for which patterns count as knowledge.

Neighbouring Fields

Statistics

Inference from samples: what may be concluded, and with what uncertainty.

Machine Learning

Learning from past experience to improve at a task in the future.

Artificial Intelligence

Building systems that behave intelligently.

Data Engineering

The twin field: managing, organising and providing access to data.

Data science is not a subset of any of them. It is the discipline of getting from a **domain question** to a **defensible answer**, using whichever of these tools fits.

Three Kinds of Task

Explorative

Find patterns.

Which customers behave similarly?

Descriptive

Summarise what is there.

What does an average customer spend?

Predictive

Anticipate future outcomes.

Is this transaction fraudulent?

Naming the task first constrains everything else — the method, the evaluation measure, and what counts as success.

Data science is defined by its **question and its domain**, not by its tools. Explorative, descriptive and predictive tasks demand different methods and different evidence — so the first decision in any project is which one you are doing.

Module 4

How Knowledge Is Acquired

> ***Guiding Question:*** *A pattern in data is not yet knowledge. What has to happen in between?*

Three Modes of Reasoning

	From	To	What it guarantees
Deduction	premises	what **must** follow	certain — but adds no new content
Induction	observed patterns	a **regularity**, a hypothesis	nothing — content-adding, not guaranteed
Abduction	an observation	the **best available explanation**	nothing — only comparable against rivals

Machine learning lives in the bottom two rows.

The Cycle, on a Real Case: Uranus

DEDUCTION

Newton's laws plus the known planets give the orbit Uranus **must** follow.

INDUCTION

Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

ABDUCTION

Best explanation: an **unseen planet** further out, pulling on Uranus.

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The cycle turns: the abduced planet becomes a new premise. Le Verrier deduces where it must be; in 1846 Galle points a telescope there and finds **Neptune**, about a degree from the prediction.

An Anomaly Leaves Two Moves

SAVE THE THEORY

Add a hidden cause

The bet: the theory is right; our picture of the world is incomplete.

If wrong: we chase something that is not there.

CHANGE THE THEORY

Replace the premises

The bet: the theory fails here, however well it worked elsewhere.

If wrong: we throw away two centuries of correct predictions.

1859: Mercury's orbit turns about **43″ per century** more than Newton allows. Which move would you make?

Same Move, Opposite Outcomes

anomaly	the move	what happened
Uranus, 1840s	a hidden planet	**Neptune** found, 1846 ✓
Mercury, 1859	a hidden planet, "Vulcan" (Le Verrier again)	?

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Vulcan is **never found** ✗ In 1915 **general relativity** explains the 43″ with no new planet — here, changing the theory was right.

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> Both decisions were **reasonable when they were made**. What separated them was evidence that did not exist yet.

Deciding Before the Evidence Is In

- → **Trust earned.** After two centuries of success, absorbing one anomaly *into* the theory was the rational first move.
- → **Evidence accumulates.** Every failed search made "a hidden planet" less plausible — confidence moved step by step, not in one jump.
- → **Replace, don't just refute.** Newton fell when a rival explained *more*, not at the first failure.
- → **Costs decide.** Chasing Vulcan forever and abandoning a good theory too early are both expensive; which move to make depends on what being wrong costs.

> A hidden planet is an **assumption added to save a pattern**. It is worth exactly as much as the check it survives.

Mini-Exercise

A dashboard shows: customers who use the mobile app **churn less**. Besides "the app keeps customers", give **one other explanation** — and one experiment that tells the two apart.

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Loyal customers are the ones who **install** the app: same correlation, opposite direction. Experiment: prompt a **random** subgroup to install it. If the app causes loyalty, their churn drops; if loyalty causes the app, it does not.

Takeaway

> A pattern is not a proof. Induction and abduction **propose**; deduction turns a proposal into a **checkable prediction**; only the check makes it knowledge. When a check fails, saving the theory with a hidden cause or changing it is a **decision made before the evidence is in**.

What Counts as Knowledge?

Knowledge discovery in databases (KDD) is the non-trivial process of identifying patterns in data that are (Fayyad et al. 1996)

- **valid** — they hold on new data, beyond the sample,
- **novel** — they were not already known,
- **potentially useful** — they can support a decision or an action,
- **ultimately understandable** — a person can make sense of them.

A pattern that fails **any** of the four is not knowledge. Most patterns a large dataset offers fail at least one.

Five Steps from Data to Knowledge

Select
selection of target data

→

Clean
preprocessing

→

Transform
transformation

→

Mine
data mining → patterns

→

Interpret
interpretation / evaluation

Fayyad et al. draw the five steps in a line, with arrows back from every step: evaluation routinely sends you back to an earlier one.

Knowledge is a pattern that is **valid, novel, potentially useful and ultimately understandable**. Data mining is one step of five; it produces patterns, and only the four criteria turn one into knowledge.

Module 5

The Process and the Craft

> ***Guiding Question:*** *Most of a data-science project is spent before any model is fitted. On what, and why does it matter?*

Step 1 — Selection of Data

Types of source

- **Structured:** relational databases, CSV tables
- **Semi-structured:** XML, JSON, web APIs
- **Unstructured:** documents, text, multimedia

Types of data

media · object-relational · spatial and spatio-temporal · graphs and networks · time series and sequences · data streams

Selection is where the **first and largest** bias enters. What was not selected cannot be discovered later.

Steps 2–3 — Preprocessing and Transformation

Cleaned

errors and noise removed — *garbage in, garbage out*

Integrated

combined with other sources, resolving conflicts

Transformed

features constructed or reduced into a representation for analysis

> These steps routinely consume most of a project's effort — and every choice in them is a modelling decision, not a clerical one.

Steps 4–5 — Mining and Interpretation, then Deployment

Data mining

Apply the algorithm — clustering, classification, rule mining — that fits the task.

Interpretation / evaluation

Test the patterns against the four criteria: valid, novel, potentially useful, ultimately understandable.

Then: deployment

Not one of Fayyad's five steps (CRISP-DM adds it): put the result where a decision is made, and monitor it there.

> The process is a **loop**, not a pipeline. Evaluation routinely sends you back to selection.

What the Work Actually Looks Like

Ask

Turn a domain question into something a dataset could answer — and check that it is the question worth answering.

Build

Acquire, clean, model, evaluate. Iterate, mostly backwards.

Communicate

Deliver a result someone can act on, with its uncertainty attached.

Three Skill Areas

Domain knowledge

What the data means, which values are impossible, which differences matter.

Statistics and modelling

What may be inferred, with what uncertainty, under which assumptions.

Engineering

Getting the data, at scale, reproducibly, into production.

The interesting failures happen **between** the three: a technically flawless model of the wrong quantity, or a correct analysis nobody can deploy.

The work runs from a **question** to a result someone can **act on**, and it spans three skill areas that nobody holds equally. The characteristic failures happen at the seams — a flawless model of the wrong quantity, or a correct analysis nobody can deploy.

Module 6

Applications, and Their Failures

> ***Guiding Question:*** *What happens to a model's validity when the world it measures changes — sometimes because of models?*

Web and Social Media

Google Flu Trends

Search queries were used to nowcast influenza faster than public health reporting.

And then it failed

It overestimated flu levels — in 2012/13 by more than double. Causes named: overfitting to seasonal search terms, **changes to Google's own search** (e.g. suggested queries), and searches driven by media coverage of the flu season (Lazer et al. 2014).

The world a model measures **keeps changing** — sometimes because of other models. Validity is not a property a model keeps once earned.

Text and Language

Classic tasks

Summarisation, entity extraction, sentiment analysis, question answering — each once a research programme of its own.

Today

A single pretrained model handles all of them, with the task specified in words rather than in code.

What did **not** change: the need to evaluate on data the model has not seen, and to state what the output may be used for.

Business and the Public Sector

Reported potentials from the era of the first big-data wave: large retail revenue gains, hundreds of billions in healthcare savings, comparable figures for e-government.

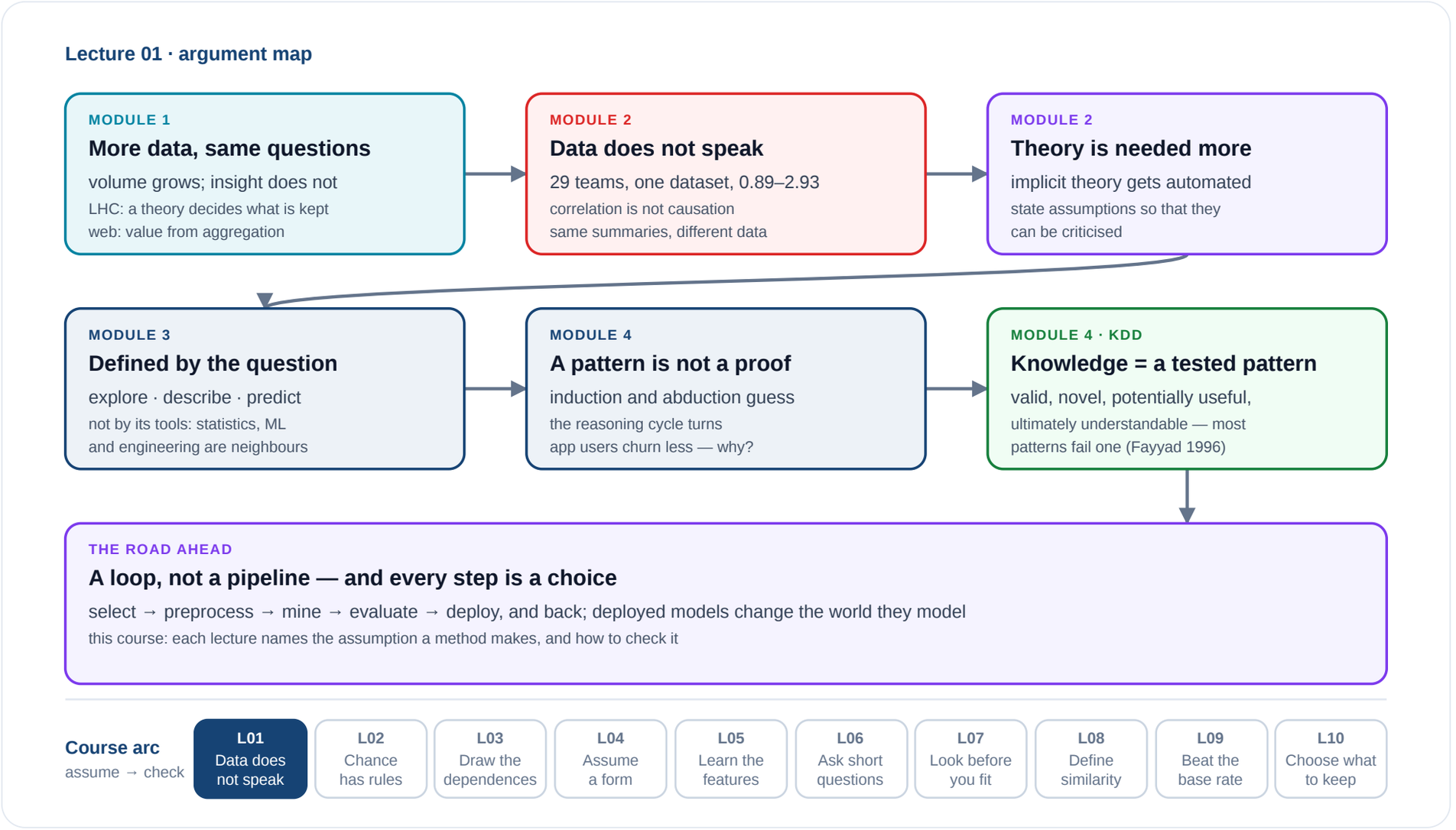
These are **projections from consultancy reports**, not measured outcomes. Treat them as claims to be interrogated — whose baseline, whose costs, and what happened afterwards?

Applications show what is possible; their failures show what must be checked. Ask of every deployed model: what changed in the world because it exists?

The Road Ahead

> ***Guiding Question:*** *Every lecture from here on brings a method. What should you ask of each one before you believe its patterns?*

Argument Map



The Road Ahead: Name the Assumption, Then Check It

L	The assumption that speaks for the data	The check
02 Chance has rules	a model of chance: outcomes, probabilities	would chance alone produce this?
03 Draw the dependences	a graph: a missing edge claims independence	does the data fit the factorisation?
04 Assume a form	a line, log-odds, independent attributes	error on unseen data, against a baseline
05 Learn the features	stacked units learn form and features	training vs validation error
06 Ask short questions	short trees first; a majority vote	pruning on held-out data
07 Look before you fit	a scale per column; why values are missing	plots, quality checks
08 Define similarity	a distance, a linkage, k	cluster validation
09 Beat the base rate	co-occurrence counts, thresholds	lift against independence
10 Choose what to keep	variance, distances or co-occurrence	explained variance, stress, biplot

> Knowing a method's assumption tells you where its patterns stop being knowledge.

The Lectures on the KDD Loop

Select	→	Preprocess	→	Transform	→	Mine	→	Interpret
L07		L07		L07 · L10		L03–L06 · L08 · L09		L02 · L04 · L08 · L09

The course does not walk the loop in order. Lecture 02 builds the language every check is written in, and Lectures 03–06 build models first, because judging data needs both. Lecture 07 then returns to the first steps, before the methods without labels (08–10).

Image Credits & References

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